

HEALTHCONNECT INITIAL ANALYSIS REPORT

1. Executive Summary

HealthConnect Clinic is experiencing operational challenges associated with missed patient appointments, including difficulty understanding the factors associated with no-shows and inefficient utilisation of appointment slots.

As part of the Data Analytics track, this initial analysis focuses on understanding the appointment dataset, assessing its quality, identifying variables relevant to attendance and no-shows, defining business questions, proposing meaningful KPIs, and establishing an analytical approach for subsequent project stages.

The dataset contains 5,000 appointment records across 18 fields covering patient demographics, appointment characteristics, booking behaviour, previous appointment history, reminder activity, distance to the clinic, waiting time, and appointment outcomes.

Initial assessment identified 2,423 No-Show appointments, 2,314 Attended appointments, and 263 Cancelled appointments. No-show appointments therefore represent 48.46% of all records. When cancelled appointments are excluded, no-shows account for approximately 51.15% of appointments with a final attended/no-show outcome.

2. Business Context

The central project question is:

How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?

The Data Analytics track contributes to this objective by establishing an evidence-based understanding of appointment attendance and identifying patterns that can support operational decision-making.

3. Analytical Objective

The objective of this analysis is to:

1. Understand the structure and suitability of the HealthConnect appointment dataset.
2. Assess the quality and consistency of the available data.
3. Identify variables that may be relevant to appointment attendance and no-shows.
4. Define business questions that can be investigated using the available data.
5. Propose meaningful KPIs for subsequent analysis.
6. Establish an initial analytical methodology for Week 5 and later project stages.

4. Data Source and Dataset Overview

The primary analytical resource is: HealthConnect_Appointment_Data.csv.

Dataset profile

Attribute	Assessment
Records	5,000
Variables	18
Unique patients	1,696
Duplicate complete records	0
Primary outcome field	appointment_outcome
Date fields	booking_date, appointment_date

5. Variable Assessment

The available fields can be grouped into the following analytical domains.

Patient characteristics

- patient_id
- gender
- age
- age_group

These variables support analysis of whether attendance patterns differ across patient demographic groups.

Appointment characteristics

- appointment_id
- appointment_type
- appointment_date
- appointment_day
- appointment_time

These variables allow appointment patterns to be assessed by service type, day and time.

Booking behaviour

- booking_date
- booking_lead_days

These fields can be used to investigate whether the amount of time between booking and appointment is associated with attendance.

Previous attendance behaviour

- previous_appointments
- previous_no_shows

These are particularly important because historical attendance behaviour may provide insight into repeat no-show patterns.

Reminder activity

- reminder_sent
- reminder_channel

These fields allow investigation of reminder coverage and differences in attendance outcomes by reminder channel.

Accessibility and service experience

- distance_to_clinic_km
- waiting_time_minutes

These variables provide an opportunity to investigate whether accessibility and waiting experience are associated with attendance.

Appointment outcome

- appointment_outcome

This is the principal outcome field and contains:

- Attended
- No-Show
- Cancelled

The outcome field is central to the proposed analysis because the Data Analytics track is specifically concerned with appointment attendance and no-show patterns.

6. Initial Data Quality Assessment

The dataset was reviewed using Python and Pandas to assess completeness, duplicates, data types, and logical consistency.

Duplicate Records

No duplicate records were identified.

Assessment: No duplication issue was identified.

Missing Values

Missing values were identified in three variables:

Variable	Missing Records	Missing %
reminder_channel	1,366	27.32%
distance_to_clinic_km	90	1.80%
waiting_time_minutes	60	1.20%

Interpretation

The missing reminder_channel values require careful interpretation.

The dataset indicates that records where reminder_sent = No have no reminder channel. Therefore, these missing values appear to be structurally related to the reminder process, rather than automatically representing data-entry errors.

This distinction will be maintained during subsequent analysis.

The missing distance and waiting-time values require further assessment before deciding whether they should be excluded, imputed, or retained as missing.

7. Data Validation

Several logical consistency checks were performed.

Booking and appointment dates

The analysis confirmed that booking dates do not occur after appointment dates.

Booking lead time

booking_lead_days was compared with the difference between appointment_date and booking_date.

No inconsistencies were identified.

Appointment day

The recorded appointment_day was compared against the weekday derived from appointment_date.

No inconsistencies were identified.

Previous no-shows

The analysis confirmed that:

$\text{previous_no_shows} \leq \text{previous_appointments}$

No violations were identified.

Reminder consistency

Records where reminder_sent = No were checked for an associated reminder channel.

No inconsistent reminder-channel assignments were identified.

Overall assessment: The dataset is sufficiently structured for the proposed initial analytical work, subject to the treatment of identified missing values and confirmation of business definitions.

8. Initial Outcome Profile

The observed appointment outcomes are:

Appointment Outcome	Records	Share
No-Show	2,423	48.46%
Attended	2,314	46.28%
Cancelled	263	5.26%
Total	5,000	100%

The volume of no-shows is therefore a significant analytical concern.

When cancelled appointments are removed from the denominator, the observed no-show proportion among appointments resulting in either attendance or no-show is approximately 51.15%.

This establishes a strong business case for investigating the characteristics associated with missed appointments.

9. Initial Analytical Observations

These observations are directional findings for further investigation, and not final conclusions.

Previous no-show history

The data indicates an increasing pattern in no-show rates as historical no-show counts increase.

Among appointments with an outcome of either Attended or No-Show:

Previous No-Shows	Records
0	2,921
1	1,548
2	438
3	78
4	12
5	3

The pattern suggests that previous attendance behaviour should be treated as an important analytical dimension.

Reminder status

Among non-cancelled appointments:

Reminder Sent	Records	No-Show Rate
No	1,285	54.63%
Yes	3,452	49.86%

There is an observed difference in no-show rates between appointments with and without reminders.

However, this does not establish that reminders cause lower no-show rates. Reminder allocation may itself be influenced by patient characteristics or operational procedures.

Therefore, the Week 5 analysis should investigate reminder status and channel while clearly distinguishing association from causation.

10. Business Questions

Based on the dataset structure and HealthConnect's business problem, the following questions are proposed for subsequent analysis.

- Overall attendance

What proportion of scheduled appointments are attended, cancelled and missed, and how does the no-show rate vary over time?

- Historical patient behaviour

Which patient characteristics and previous appointment behaviours are associated with higher no-show rates?

- Appointment characteristics

How do appointment type, appointment day and appointment time relate to attendance outcomes?

- Reminder activity

How does appointment attendance vary according to reminder status and reminder channel?

- Accessibility and waiting experience

Are distance to the clinic and waiting time associated with appointment no-shows?

These questions provide a focused analytical framework without expanding the scope beyond what can reasonably be supported by the available data.

11. Proposed KPIs

I propose the following five KPIs.

KPI	Purpose
No-Show Rate	Measures the scale of missed appointments
Attendance Rate	Measures successful utilisation of scheduled appointments
Cancellation Rate	Quantifies appointments cancelled before attendance
Repeat No-Show Rate	Measures recurring missed-appointment behaviour
Reminder Coverage Rate	Measures the proportion of appointments receiving reminders

KPI definitions

No-Show Rate

$\text{No-Show appointments} / \text{eligible scheduled appointments} \times 100$

Attendance Rate

$\text{Attended appointments} / \text{eligible scheduled appointments} \times 100$

Cancellation Rate

$\text{Cancelled appointments} / \text{total scheduled appointments} \times 100$

Repeat No-Show Rate

$\text{No-show appointments among patients with previous no-show history} / \text{eligible appointments} \times 100$

Reminder Coverage Rate

Appointments where a reminder was sent / total scheduled appointments × 100

12. Initial Analysis Approach

The proposed analytical process for the next stage is:

Data Validation



Outcome Distribution



Temporal Analysis



Patient Characteristics



Previous Attendance Behaviour



Appointment Type / Day / Time



Booking Lead Time



Reminder Status & Channel



Distance & Waiting Time



Cross-Dimensional Analysis



Identify Meaningful Patterns



Business Insights



Actionable Recommendations

The analysis will focus on identifying patterns and associations rather than assuming causality.

13. Assumptions

The following assumptions apply to the initial analysis:

1. Each row represents one scheduled appointment.
2. appointment_outcome represents the final recorded outcome of the appointment.
3. booking_lead_days represents the number of days between booking and appointment.
4. previous_no_shows represents historical no-show behaviour before the current appointment.
5. A missing reminder_channel where reminder_sent = No is considered structurally expected.
6. No causal relationship will be inferred solely from observed differences in the dataset.
7. Small subgroups will be interpreted cautiously.

14. Risks and Dependencies

Risk / Dependency	Potential Impact	Proposed Response
Missing distance/waiting-time data	May affect segment-level analysis	Assess missingness before analysis
Reminder-channel missing values	Could be incorrectly treated as data errors	Interpret alongside reminder_sent
Small subgroups	Risk of misleading conclusions	Report sample sizes and aggregate where appropriate
Correlation interpreted as causation	Could lead to inappropriate recommendations	Clearly distinguish association from causality
Unconfirmed business definitions	Could affect KPI calculations	Validate against Data Dictionary
Fictional dataset	Findings may not represent real clinic behaviour	Present conclusions within project context

15. Proposed Week 5 Focus

The proposed focus is:

1. Deeper exploratory analysis

Investigate no-show patterns across:

- Age groups
- Gender
- Appointment type
- Appointment day
- Appointment time
- Booking lead time
- Previous no-shows
- Reminder status
- Reminder channel
- Distance to clinic
- Waiting time

2. KPI development

Operationalise the proposed KPIs after confirming their definitions and denominators.

3. Comparative analysis

Identify the strongest observable patterns associated with no-shows and attendance.

4. Business interpretation

Translate analytical findings into operational implications for HealthConnect.

5. Reporting foundation

Begin developing the analytical outputs that can later support the project's dashboard and final recommendations.

16. Conclusion

The initial assessment establishes that the HealthConnect appointment dataset is sufficiently structured to support the proposed Data Analytics work.

The most significant initial observation is the high volume of missed appointments, with 2,423 of 5,000 records classified as No-Show. Historical no-show behaviour also presents a potentially important analytical relationship, while reminder activity, appointment characteristics, booking lead time, distance, and waiting time provide additional dimensions for investigation.

The analysis has also established that the dataset requires careful interpretation rather than indiscriminate cleaning. In particular, missing reminder channels appear to be structurally associated with appointments where no reminder was sent, while missing distance and waiting-time values require further assessment.